MIL-T-23445A(SHIPS)
22 June 1966
SUPERSEDING
MIL-T-23445(SHIPS)
(See 6.4)

MILITARY SPECIFICATION

TEST UNIT FOR CHECKING INTEGRITY OF FUEL FILTER-SEPARATOR ELEMENTS

1. SCOPE

1.1 This specification covers a test unit for checking the integrity of coalescer and separator elements furnished in accordance with MIL-F-15618 before their installation into a filter-separator.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
P-D-680 - Dry Cleaning Solvent.

MILITARY
MIL-D-963 - Drawings, Electrical Hull and Mechanical Equipment for Naval Shipboard Use.
MIL-G-5572 - Gasoline, Aviation: Grades 80/87, 91/96, 100/130, 115/145.
MIL-J-5624 - Jet Fuel, Grades JP-4 and JP-5.
MIL-T-15071 - Manuals, Equipment and Systems.
MIL-P-15137 - Provisioning Technical Documentation for Repair Parts for Electrical and Mechanical Equipment (Naval Shipboard Use).
MIL-F-15618 - Filter-Separators and Filter Elements, Fluid, Pressure, Aviation and Diesel Fuel, Naval Shipboard.

STANDARD

MILITARY
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-761 - Electric Power, Alternating Current for Shipboard Use, Characterization and Utilization of.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Ratings, Rules and Regulations.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York, N.Y. 10016.)

3. REQUIREMENTS

3.1 Preproduction unit. - As soon as practicable, after approval of the preliminary drawings and prior to the beginning of production, the contractor shall furnish a preproduction unit for the examination and tests specified in 4.4 and 4.5 to determine compliance with the requirements of this specification (see 6.3).

FSC 4330

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3.1.1 Production of units prior to approval of the preproduction unit shall be at the contractor's risk. Approval of the preproduction unit does not relieve the contractor of his obligations to supply equipment conforming to this specification. Deviations from the approved design are not acceptable without prior written approval from the command or agency concerned.

3.2 Materials.- All materials used in the construction of the test unit shall be resistant to sea water corrosion and shall be compatible with grade JP-5 fuel of MIL-J-5624, type II solvent of P-D-680 and aviation gasoline of MIL-G-5572. Copper-nickel alloy (70-30 and 90-10) are acceptable as well as brass, bronze and copper. Brass or bronze, if used, shall not contain more than 16 percent zinc. Aluminum is acceptable for use only in the frame, base, and similar structural applications provided it does not come in contact with copper base nonferrous alloys. The selection of materials used shall be subject to approval of the command or agency concerned. The complete unit, with the exception of the tank and piping interior, mating working surfaces, and identification tags, shall be finished with a coating of zinc chromate primer.

3.3 General requirements.- This element test unit shall be a simply constructed, ruggedly designed apparatus to test the integrity of both coalescer and separator elements before installation into a fuel filter-separator. The unit shall have two systems; an emulsion producing system for testing coalescer elements and an air system for testing separator elements (see figure 1). Figure 1 is to be used merely as a design guide and deviations which will not affect the intended application may be made subject to the approval of the command or agency concerned. The unit shall be portable and easily handled by two men (see 3.6). Means shall be provided to permit draining the test unit of all fuel and water without requiring any dismantling of equipment or breaking of connections.

3.4 Interchangeability.- The unit shall be capable of testing the integrity of all sizes and types of coalescer and separator elements furnished in accordance with MIL-F-15608. The unit shall also be capable of testing the integrity of permanent type separator elements. The overall physical dimensions and construction of these elements are the same as the paper elements specified in MIL-F-15618 except that the hydropholic media of the permanent type element is a 200 by 200 mesh monel wire screen (0.0021 inch diameter) coated on both sides with polytetrafluoroethylene. The hydropholic media has an inner diameter, after forming, of not less than 3-3/8 inches.

3.5 Dimensions.- The completely assembled test unit shall not exceed the following maximum dimensions:

(a) Width - 20 inches
(b) Length - 50 inches
(c) Height - 36 inches

3.6 Weight.- The dry weight of the complete test unit shall be as low as practicable and shall not exceed 230 pounds since the unit is to be operated as a portable unit. The weight of the finished product will be an important factor in making the award.

3.7 Performance requirements.- The test unit shall accomplish two types of tests; namely, "Emulsion" and "Air Bubble" tests. The system required to perform each of these tests shall function independently of the other and shall consist essentially of the components specified in 3.7.1 to 3.7.1.6 and 3.7.2. Any variations shall be as approved by the command or agency concerned.

3.7.1 Emulsion test system for checking coalescer elements.- A system as specified herein shall be provided to check the condition of coalescer elements. The flow through the element shall be in the inside-to-outside direction regardless of the element being tested.

3.7.1.1 Pump.- The pump shall be a standard commercial centifugal pump suitable for the purpose. It shall have a capacity of approximately 15 gallons per minute (g.p.m.) at a discharge pressure of about 8 pounds per square inch (p.s.i.). The arrangement of components shall be such that the pump will be required to have a minimum suction head.

3.7.1.2 Motor.- The motor shall be a standard commercial electric motor of explosionproof construction. The motor shall be 110 volt, 60 cycle, single phase, or 440 volt, 60 cycle, 3 phase.

3.7.1.3 Flow indicator.- A standard commercial sight flow fitting shall be installed in the discharge line from the pump and in a position to be easily observed by the operator.

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3.7.1.4 Pressure gage. - A pressure gage shall be supplied to indicate the outlet pressure of the pump. The range of the gage shall be 0-15 p.s.i.

3.7.1.5 Sampling connection. - A sampling connection shall be supplied as close to the filter mount inlet as possible to continuously check the operation of the unit.

3.7.1.6 Piping. - The piping shall be arranged to permit the depth of the suction line to the pump to be adjusted in the tank as necessary by the operator, so that a mixture of fuel and water is pumped. Accomplishment of this feature by permanently installed piping will be considered also.

3.7.2 Air bubble test system for checking separator elements. - A motor operated air compressor shall be used to supply air to the interior of the separator element. The motor shall be of explosionproof construction and operable from shipboard 115 volt, 60 cycle, single phase power supply or 440 volt, 60 cycle, 3 phase power supply as specified in MIL-STD-761. The air system shall be as shown in figure 1 and shall include the following:

(a) A regulating valve capable of providing the required regulated pressure when operating between lockup and the maximum required flow.
(b) A supply restriction (needle valve with a provision for locking the setting or a properly sized orifice) downstream of the regulator and a needle valve downstream of the supply restriction venting to atmosphere. This system shall be capable of providing smooth and accurate continuous bleed control of the pressure to the element between the range of 2 to 20 inches water by manipulation of the vent valve. The following control requirements shall be met:
  (1) The system shall be sized so that under any steady state operating condition, breakthrough at the element will not drop the regulated pressure to the element, as monitored by the manometer, by more than 10 percent.
  (2) The vent needle valve shall be sized, in relation to the supply restriction, so that under any operating condition, 75 percent of its range is useable for the control of the regulated pressure from 2 to 20 inches of water.
(c) A 30-inch U-type water manometer shall be used to indicate the pressure within the element.

3.7.2.1 Due to element design, evidence of element damage should show up at air pressures below 6 inches of water for the paper type elements and below 4 inches of water for the permanent type separator element. A sound paper type element should show an overall bubble breakthrough between 8 and 15 inches of water pressure.

3.7.3 Additional components. - Additional components shall be supplied as specified in 3.7.3.1, 3.7.3.2 and 3.7.3.3.

3.7.3.1 Tank. - The tank shall be a horizontal, open top tank. It may be of sufficient length to permit changing elements without disassembling the tank, or may be shorter provided that this will not increase the element changing time specified in 3.8. Portions of the tank as necessary shall be constructed of transparent shock resistant material (no glass) to provide good visibility of the element to the operator. A vortex breaker shall be provided on suction pipes as necessary for proper operation.

3.7.3.2 Tie-down. - The test unit shall be provided with wheels or casters capable of being locked to prevent shifting of the unit due to motion of the ship deck. Suitable lugs shall be installed also to permit the attachment of tie-down lines for securing the unit to the deck.

3.7.3.3 Drainage connection. - A drainage connection shall be provided at the lowest point of the system to permit total drainage.

3.8 Testing time. - With reasonable handling, it shall take one man (operator) no more than 30 seconds to change an element while conducting either "Emulsion" or "Bubble" test. It shall take no more than 1 minute to change over the unit from conducting the bubble test to conducting the emulsion test or vice versa. It shall take no more than 5 minutes to change from testing one size element to testing another size element (example: 17-1/2 inch long element to 24 inch long element).

3.9 Arrangement of components. - All gages, indicators, valves, and other components shall be readily accessible to and readable by the operator of the unit from the normal operating position. It shall not take more than one man to efficiently operate the test unit. All parts necessary for operation shall be identified by clear, permanent markings such as label tags, stencil or other suitable means.

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3.10 Element mount.- The element mount shall be similar in construction to figure 1 of MIL-F-15618 except that, where a reduction in tank size can be realized and a suitable mounting means provided, the center tube may be omitted. The element mount may be pivoted in order to facilitate replacement of elements. The element mount shall be so situated in the tank that when any size element is being tested the entire element shall be submerged at least 1 inch below the surface of the fuel. It shall be possible to rotate the element being tested during the test without interruption of the test and fuel or air shall not bypass the element at the sealing surfaces while the element is being rotated. Removal of the element from the unit after testing shall be accomplished without spillage of the test fluid outside the tank.

3.11 Drawings.- Unless drawing requirements are waived (see 3.15), preliminary drawings shall be submitted to the command or agency concerned for approval prior to manufacture of the production unit. Final drawings, validated by the Government inspector, shall be furnished showing the design of the production units supplied. Drawings shall contain sufficient details of components used as well as assemblies and sectional views to convey a clear understanding of the unit, installation, and mounting information. A list of materials for parts shall be included. The materials used shall be identified by a Government or commercial specification and type, where applicable. Drawings prepared in accordance with good commercial practice are acceptable. MIL-D-963 may be used as a guide.

3.11.1 Certification data sheets shall be furnished in accordance with MIL-D-963.

3.12 Manuals.- Unless manual requirements are waived (see 3.15), manuals shall be furnished in accordance with type I of MIL-M-15071 in the quantities specified (see 6.2). Each test unit shall have two copies of the manual enclosed. Four copies of the final manual shall be forwarded to the cognizant command or agency concerned unless otherwise specified in the contract or order.

3.13 Repair parts and special tools.- Repair parts which are determined to be necessary for maintenance support of the test units shall be established in accordance with MIL-P-15137. Repair parts shall be furnished only when specified (see 6.2.). The Supply Demand Control Point for units under the Naval Ship Engineering Center cognizance is Ships Parts Control Center, Mechanicsburg, Pa. The need for special tools to use the test unit shall be avoided if at all possible. If not possible, the special tools shall be supplied with the unit in a tool box attached to the unit. Special tools are defined as those tools not listed in the Federal Supply Catalog (copies of this Catalog may be consulted in the office of the Government inspector).

3.14 Marking.- Each test unit shall contain clearly marked permanent identification containing at least the following:

(a) Specification number
(b) Contract or order number
(c) Manufacturer's name or trade mark

3.15 Waiver of drawing and manual requirements.- When a contractor has previously furnished test units identical to those specified herein, which have been approved and unconditionally accepted by the command or agency concerned, the requirements for preliminary drawings and preliminary manuals specified herein may be waived by the command or agency concerned.

3.16 Workmanship.- Workmanship shall be first class in every respect. The operation unit shall be free of defects affecting its operating or reliability for continued use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction inspection.- Preproduction inspection shall consist of the examination of 4.4 and the tests of 4.5. Failure of the preproduction unit in any examination or test shall be cause for rejection and the command or agency concerned shall be notified as soon as possible. A complete certified test report including test procedures and results shall be forwarded to the command or agency concerned.

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4.3 Quality conformance inspection. -

4.3.1 Lot. - A lot shall consist of all testing units manufactured under essentially the same conditions and offered for delivery at one time.

4.3.2 Sampling for examination and tests. - Not less than 10 percent (to nearest higher whole number) of the lots shall be selected at random for the examination and tests specified in 4.4 and 4.5. If any unit fails any examination or test the entire lot shall not be offered for delivery.

4.4 Examination. - Test units shall be examined to verify conformance to all the requirements of this specification not involving tests. In addition, all parts intended to move shall be checked to insure that they are securely attached and functioning properly. Each assembled testing unit shall be weighed (dry).

4.5 Tests. -

4.5.1 Test fluid. - The test fluid used for testing elements for JP-5 service shall be grade JP-5 in accordance with MIL-J-5624. The test fluid for testing elements for gasoline service shall be solvent in accordance with type II of P-D-680.

4.5.2 Coalescer element test. -

4.5.2.1 Emulsion test. - A coalescer element in accordance with MIL-F-15618 shall be installed in the test units. The tank shall be filled with the correct amount of JP-5 fuel and water. The system shall be started up and after 2, 5, and 10 minutes of running time, a one-liter emulsion sample shall be taken in a graduated one-liter bottle at the sampling connection to check emulsion quality. The sample shall have a hazy appearance and shall contain no less than 5 percent water but no more than 15 percent water by volume. It shall take at least 10 minutes for the haze to settle. After the water has settled out of the sample the percentage shall be determined by checking the graduations on the sample bottle.

4.5.2.2 Element leakage test. - Each size coalescer element shall be rotated on the element mount with the pump in operation. There shall be no bypassing of the emulsion around the element end seals while rotating the element or the element mount or while the element is stationary on the mount. This test shall be conducted with all the different sized coalescer elements listed in MIL-F-15618. A second element will form large drops of coalesced water over the element surface. A defective element is indicated by one or more cloudy streams issuing from the defective portion.

4.5.3 Separator element test. -

4.5.3.1 Bubble system test. - A separator element in accordance with MIL-F-15618 shall be installed in the test unit. The tank shall be filled with the correct amount of JP-5 fuel and water. The emulsion system shall be secured and the bubble system shall be "lined up" for the bubble test. The air pump shall be actuated and the operator shall be able to increase the pressure to the separator until air bubbles begin to "break out" over the whole area of the separator element.

4.5.3.1.1 A satisfactory (undamaged) permanent type separator element (see 3.4) of each size (24 and 17-1/2 inch) shall be installed and tested on the test stand. These elements shall be prewetted, installed on the element mount and submerged horizontal to and 1 inch below the JP-5 fuel surface. Air pressure to the interior of the element shall be controlled by manipulating the vent needle valve. The air pressure within the element shall be slowly increased until a stream of air bubbles is observed escaping from several locations on the hydropholic media. The pressure shall be reduced at a rate of approximately 1/2 inch of water per second to 4 inches of water. The stream of bubbles escaping from the hydropholic media shall have completely ceased at 4 inches of water pressure.

4.5.3.2 Element leakage tests. - Each size separator element shall be rotated on the element mount with air pressure being supplied to the element. There shall be no bypassing of the air around or at the element end seal while rotating the element on the element mount or while it is stationary on the mount. These tests shall be conducted with all the different sized separator elements listed in MIL-F-15618.

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5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for storage of on board repair parts. -

5.1.1 Test units and repair parts. -

5.1.1.1 Preservation and packaging. - Preservation and packaging which may be the suppliers commercial practice, shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.1.2 Packing. - Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Ratings, Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the supplier's commercial practice.

5.1.1.3 Marking. - Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with MIL-STD-129. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.1.2 Manuals. -

5.1.2.1 Preservation and packaging. - Preservation and packaging shall be sufficient to afford complete protection against physical damage or deterioration during shipment and storage. Each manual shall be enclosed in a waterproof container (a suitable sealed plastic film is satisfactory for waterproofing).

5.1.2.2 Packing. - The required number of manuals to be furnished with the equipment shall be packed with the equipment in an appropriate manner so as to prevent their being damaged. Bulk manuals shall be packed for shipment in accordance with MIL-M-15071 unless otherwise specified. Prominent notation shall be made to indicate that manuals are enclosed if the manuals are not readily visible.

5.1.2.3 Marking. - Bulk shipments of manuals shall be marked in accordance with MIL-M-15071.

5.2 Domestic shipment and storage or overseas shipment. - The requirements and levels of preservation, packaging, packing and marking shall be specified by the procuring activity (see 6.2).

(5.2.1 The following provides various levels of protection during overseas shipment or long time storage, which may be required when procurement is made.

5.2.1.1 Repair parts and special tools. -

5.2.1.1.1 Preservation. - If overseas shipment or long time storage is planned, repair parts and special tools shall be protected by at least the barrier material in accordance with MIL-B-131.

5.2.1.1.2 Packaging and packing. - If overseas shipment or long time storage is planned, packing shall be level A or B as specified, in accordance with MIL-R-196.

5.2.1.1.3 Marking. - In addition to any special marking required, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.)

6. NOTES

6.1 Intended use. - Test units furnished under this specification are intended for use by ships, shipyards and other shore activities in verifying the integrity of filter-separator elements before they are placed into a filter-separator. This is highly desirable inasmuch as some elements have become damaged in shipment.

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6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Quantities of manuals required (see 3.12).
(c) If repair parts are required (see 3.13).
(d) If preservation, packaging and packing other than as specified in 5.1 is required (see 5.2).

6.3 Preproduction. - Invitations for bids should provide that the Government reserves the right to waive the requirement for preproduction samples as to those bidders offering a product which has been previously procured or tested and unconditionally accepted by the Government, and that bidders offering such products, who wish to rely on such previous production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.4 CHANGES FROM PREVIOUS ISSUE. THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

Preparing activity:
Navy - SH
(Project 4330-N026SH)

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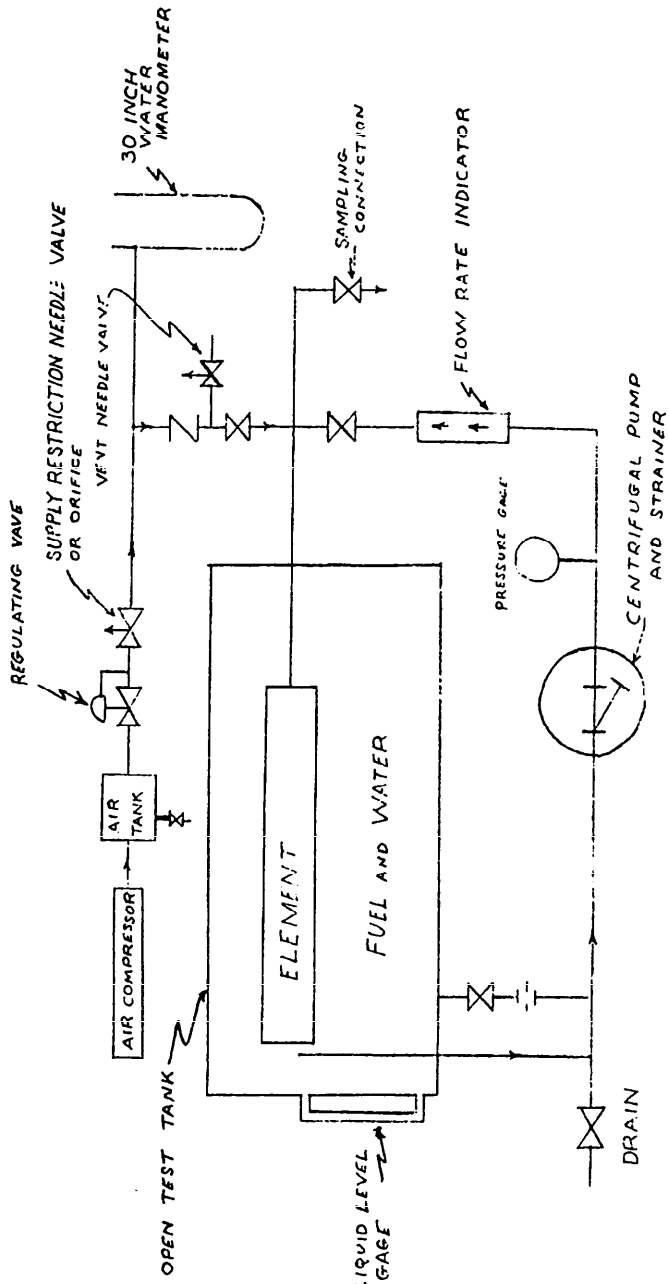


Figure 1 - Flow diagram.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD, FORM 1 APR 63 1426

REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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